

Task 1 : Classification

Objective:

This project challenges you to develop a robust machine learning classification model to predict whether a retail transaction is fraudulent. Using a dataset containing transaction details, customer activity patterns, payment information, device and location data, merchant categories, and fraud-related behavioral indicators, the primary goal is to accurately classify the target label `fraud_flag`, where 1 signifies “Fraudulent” and 0 signifies “Legitimate”.

Process Overview:

Success in this project depends on a structured machine learning workflow that includes data exploration, preprocessing, feature engineering, model development, and evaluation.

1. **Data Exploration & Understanding:** Begin by thoroughly analyzing the retail transaction dataset. Investigate the relationships between features and the target variable `fraud_flag`. The goal is to understand which customer behaviors, transaction patterns, and contextual signals are most associated with fraudulent activity.
2. **Preprocessing & Feature Engineering:** Effective data preparation is crucial for building a high-performing model. Apply appropriate techniques such as handling missing values, encoding categorical features, scaling numerical data, managing date information, and potentially engineering new features to capture predictive signals better.
3. **Data Leakage:** One of the most important points in this task is avoiding Data Leakage. Some features may introduce future information into the model if they are not calculated correctly and strictly based only on information available before the transaction time. If leakage is used and the model achieves an unrealistically high score, the result will not be accepted and that part will receive a score of zero. Therefore, participants must be very careful during preprocessing, feature engineering, and validation to ensure that no future information or validation/test information is used during training.
4. **Model Development & Selection:** Experiment with various binary classification algorithms suitable for this task. Compare their performance using appropriate validation strategies and select the model that yields the highest **F1 Score** on your validation set.
5. **Class Imbalance cautions:** In fraud detection problems, the class distribution is usually imbalanced; in other words, fraudulent transactions are often much less frequent than legitimate transactions. Therefore, participants should pay close attention to this issue during data analysis and model evaluation. In such cases, relying only on Accuracy can be misleading, because a model may achieve a high

Accuracy by mostly predicting the majority class while still performing poorly in detecting fraudulent transactions. Therefore, the impact of class imbalance on model performance should be carefully considered during model development.

6. **Classification threshold:** Many classification models produce probability scores. For example, a model may predict that a transaction has a fraud probability of 0.73. To convert this probability into a final class label of 0 or 1, a threshold is applied, and the default threshold is usually 0.5. However, in fraud detection, a threshold of 0.5 may not be optimal. Participants can test different threshold values on the validation set and choose the one that improves model performance.
7. **Comprehensive Evaluation:** While the F1 Score is the primary metric for final scoring, The recommended classification metrics to report are: Accuracy, Precision, Recall or Sensitivity, F1 Score, Confusion Matrix, ROC-AUC, Specificity, Matthews Correlation Coefficient, Cohen's Kappa if the model produces probability outputs.

Prediction & Submission:

Your final, trained model will be used to predict the fraud_flag for a separate test dataset where the true labels are withheld.

- **Output:** Submit your predictions as a **CSV file**.
- **Format:** The CSV must contain exactly two columns:
 - id: The unique identifier for each test instance.
 - label: Your model's predicted value (0 or 1).

To participate in the competition related to this task Click on the competition link: [Quera Competition](#).

Good luck with the competition!